

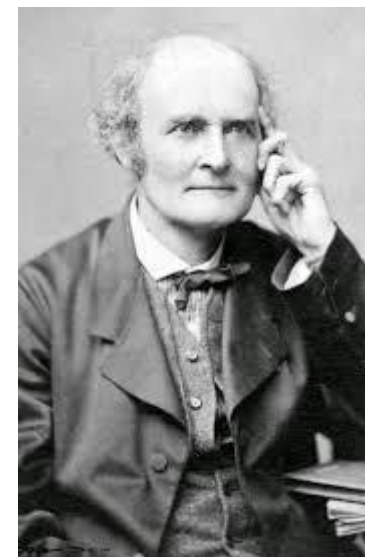
Math Primer

(9. Matrices)

Contents

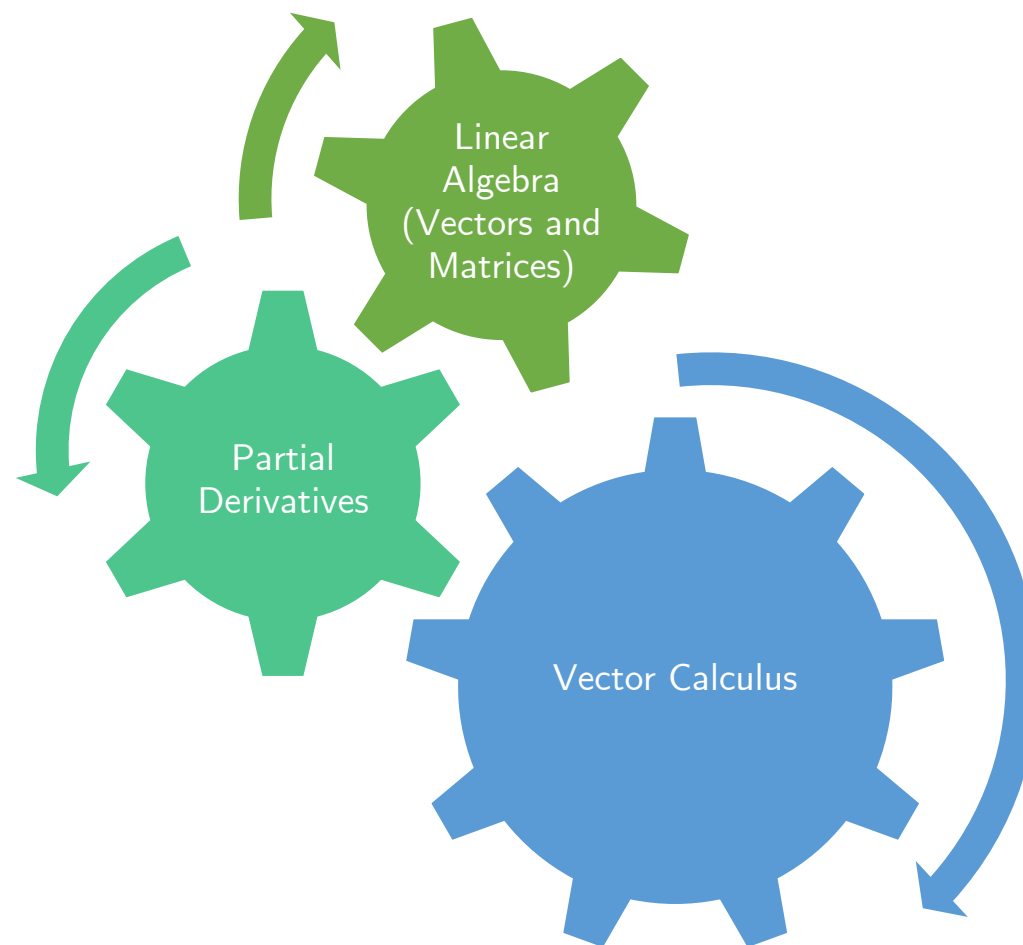


- ▶ Matrix representation
- ▶ Types of matrices
- ▶ Matrix Sum, Product, Transpose, Inverse
- ▶ Determinant, Minors and Cofactors
- ▶ Special Matrices
- ▶ Row Space, Column Space, Rank
- ▶ Transforming Vector spaces
- ▶ System of Linear Equations
- ▶ Eigen Values and Eigen Vectors



Arthur Cayley
(1821 – 1895)

The Multivariable Calculus Landscape



Matrices



- Think of Matrix as stacking of vectors

Fruit Basket 1



(3 apples, 4 Oranges)

Fruit Basket 2



(4 apples, 3 Oranges)

Apples Oranges

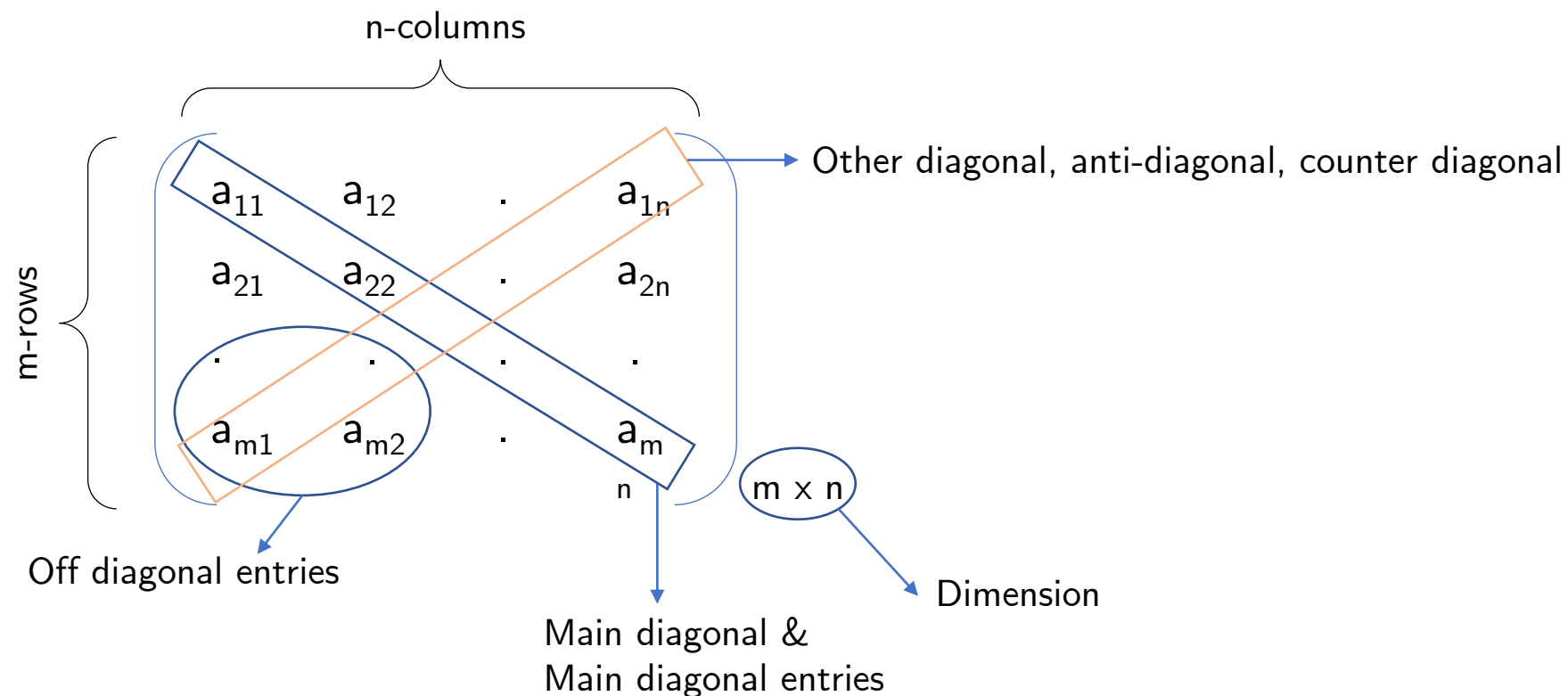
Fruit Basket 1	3	4
Fruit Basket 2	4	3

(Matrix)

- 2d matrices are most common where the need is to store data in rows and columns format. Typically, the columns remain static, the rows represent the instances or cases which are dynamic and added or deleted with ease. These are referred to as data frame in programming languages.

Matrices - Nomenclature

► Structure of a Matrix , Nomenclature...



Refer to a cell as a_{ij} . (stands for the element in i th row and j th column)

Matrices - Types

► Types of Matrices

$$\begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}_{2 \times 2}$$

Square Matrix

(no of Rows = no of columns)

$$\begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix}_{2 \times 2}$$

Diagonal Matrix

(off-diagonal entries are 0)

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}_{2 \times 2}$$

Identity Matrix

($a_{ij} = 1$ if $i = j$, 0 otherwise)

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ 3 & 1 & 3 \end{pmatrix}_{3 \times 3}$$

Symmetric Matrix

($a_{ij} = a_{ji}$)

$$\begin{pmatrix} 1 & 0 & 0 \\ 2 & 5 & 0 \\ 3 & 1 & 3 \end{pmatrix}_{3 \times 3}$$

Lower Triangular Matrix

($a_{ij} = 0$ if $j > i$)

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}_{3 \times 3}$$

Upper Triangular Matrix

($a_{ij} = 0$ if $i > j$)

$$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}_{3 \times 3}$$

Null / Zero Matrix

($a_{ij} = 0$ everywhere)

Refer to https://en.wikipedia.org/wiki/List_of_named_matrices for more details

Matrices - SUM

- ▶ To sum 2 matrices, they need to be of equal size and the corresponding elements get added
- ▶ Output is also the same dimension
- ▶ $A + B = B + A$

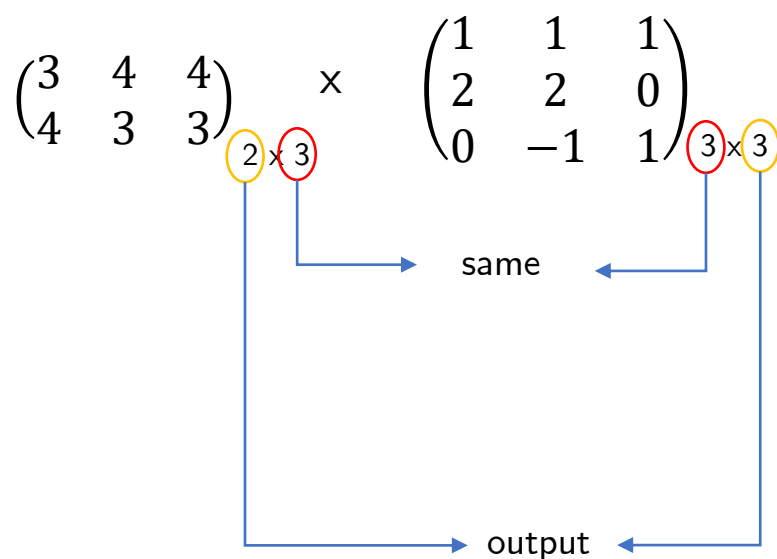
$$\begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}_{2 \times 2} + \begin{pmatrix} 0 & 1 \\ 2 & -2 \end{pmatrix}_{2 \times 2} = \begin{pmatrix} 3 & 5 \\ 6 & 1 \end{pmatrix}_{2 \times 2}$$

$$\begin{pmatrix} 3 & 4 \\ 4 & 3 \\ 4 & 3 \end{pmatrix}_{3 \times 2} + \begin{pmatrix} 0 & 4 \\ -2 & 1 \\ 3 & 1 \end{pmatrix}_{3 \times 2} = \begin{pmatrix} 3 & 8 \\ 2 & 4 \\ 7 & 4 \end{pmatrix}_{3 \times 2}$$

$$\begin{pmatrix} 3 & 4 \\ 4 & 3 \end{pmatrix}_{2 \times 2} + \begin{pmatrix} 0 & 4 \\ -2 & 1 \\ 3 & 1 \end{pmatrix}_{3 \times 2} \quad \times$$

Matrices - PRODUCT

- ▶ Two matrices, $A_{m \times n}$ and $B_{p \times q}$ can be multiplied if $n = p$, and the result will be of size $(m \times q)$
- ▶ In the resulting matrix, the cell (i, j) is the dot product of i th row of 1st matrix and j th column of 2nd matrix
- ▶ $AB \neq BA$ (not commutative)



$$= \left(\begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \begin{pmatrix} 3 \\ 4 \\ 4 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right)_{2 \times 3}$$

$$= \begin{pmatrix} 11 & 7 & 7 \\ 10 & 7 & 7 \end{pmatrix}_{2 \times 3}$$

Matrices - TRANSPOSE



- ▶ Matrix TRANSPOSE is denoted as A' or A^T . A transpose operation converts rows into columns and vice versa. Meaning the cell (i, j) in original matrix will appear in cell (j, i) of the transposed matrix.

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

$$A' = A^T = \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix}$$

- ▶ Properties of Transpose

$$A^{T^T} = A$$

$$(A + B)^T = A^T + B^T$$

$$(kA)^T = kA^T$$

$$(AB)^T = B^T A^T$$

Matrices – DETERMINANT, MINORS, COFACTORS



$$|A| = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc \quad (\text{Determinant of a } 2 \times 2 \text{ matrix})$$

$$|A| = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix} = aei + bfg + cdh - ceg - bdi - afh$$

minors

M_{11} M_{12} M_{13} M_{ij} (for cell (i,j))

cofactors

$(-1)^{1+1}M_{11}$ $(-1)^{1+2}M_{11}$ $(-1)^{1+3}M_{11}$ $(-1)^{i+j}M_{11}$

- ▶ $\det(A)$ or $|A|$ = row-wise or column-wise sum-product of elements and their co-factors
- ▶ $\text{adj}(A)$ = transpose of cofactor matrix
- ▶ $\det(AB) = \det(A) \det(B)$
- ▶ $\det(A^T) = \det(A)$
- ▶ $\det(A^{-1}) = 1/\det(A)$

Matrices – DETERMINANT, MINORS, COFACTORS



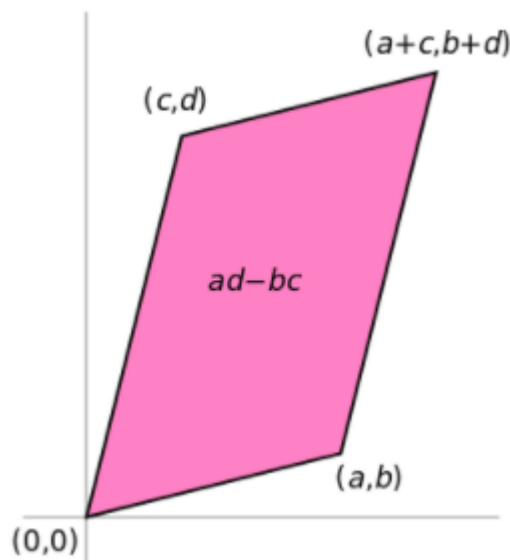
Exercise – Verify the adjoint of the matrix below.

$$\text{adj} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{bmatrix}$$

Matrices – DETERMINANT (Geometric Interpretation)

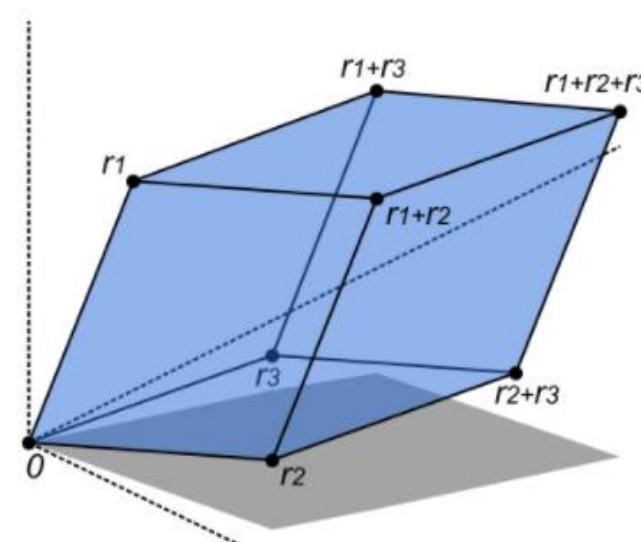


- Determinant is a scalar quantity. The geometric interpretation of determinant is that it captures the volume enclosed by the vectors



$$|A| = \begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - bc$$

Area of the parallelogram



$$|A| = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

Volume of a parallelepiped

Matrices – INVERSE

- ▶ For a matrix A to be **invertible**, it must be a square matrix and $\det(A) \neq 0$
- ▶ Inverse is given using adjoint and determinant : $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$

Properties

$$AA^{-1} = A^{-1}A = I$$

$$(kA)^{-1} = \frac{1}{k}A^{-1}$$

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$(A^T)^{-1} = (A^{-1})^T$$

$$A = \begin{pmatrix} 1 & 2 & 0.5 \\ 0.5 & -1 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

$$A^{-1} = \begin{pmatrix} 5 & 1 & -4.5 \\ -1.5 & -0.5 & 1.75 \\ -2 & 0 & 2 \end{pmatrix}$$

Matrices – Special types



► Here are some special matrices ...

Condition	Name
$A^T = A$	Symmetric
$A^T = -A$	Skew-symmetric
$A^T = A^{-1}$	Orthogonal
$A^n = A$	Idempotent
$ A = 0$	Singular
$\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$	Positive definite
$\mathbf{x}^T A \mathbf{x} \geq 0$ for all $\mathbf{x} \neq \mathbf{0}$	Positive semi-definite



Matrices – Column space, Row Space and Rank

- ▶ Recall that in a set of vectors, if no vector can be created by a linear combination of others, then they are called linearly independent.
- ▶ Another way to verify the above is if there exists a linear combination of all the vectors (where all the scalars are not 0) that leads to 0 , e.g., $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 5 \end{pmatrix}$ are dependent, because

$$\begin{pmatrix} 3 \\ 5 \end{pmatrix} = (1) \begin{pmatrix} 1 \\ 2 \end{pmatrix} + (1) \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{or} \quad (1) \begin{pmatrix} 3 \\ 5 \end{pmatrix} - (1) \begin{pmatrix} 1 \\ 2 \end{pmatrix} - (1) \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- ▶ **Column space** is the span of column vectors
- ▶ **Row space** is the span of row vectors (= column space of the transpose)
- ▶ The dimensions spanned by the column vectors is called '**rank**' of a matrix (= number of linearly independent columns). If all column vectors are independent, matrix has **full rank**.
- ▶ Row space = Column space for a square matrix
- ▶ Rank of a square matrix = number of linearly independent columns or rows
- ▶ If a square matrix is NOT full rank, then its determinant = 0

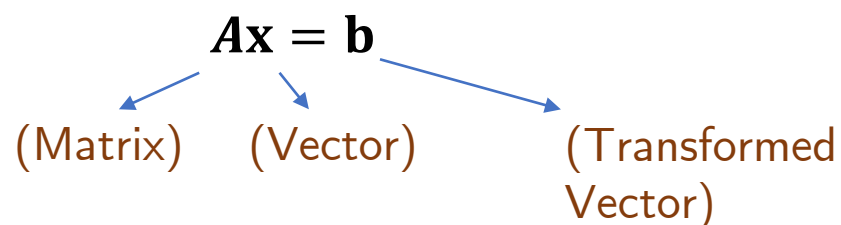
Matrices – Column space, Row Space and Rank (Examples)



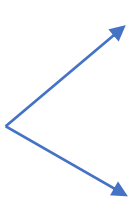
		Column space	Row space	Rank
$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$	$\longrightarrow$	whole of $\mathbb{R}^2$	whole of $\mathbb{R}^2$	2 (full rank)
$\begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$	$\longrightarrow$	whole of $\mathbb{R}^2$	2D plane in $\mathbb{R}^3$	2 (Full row rank)
$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}$	$\longrightarrow$	A line in $\mathbb{R}^2$	A line in $\mathbb{R}^2$	1
$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 4 \\ 1 & 4 & 5 \end{pmatrix}$	$\longrightarrow$	2D plane in $\mathbb{R}^3$	2D plane in $\mathbb{R}^3$	2
$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$	$\longrightarrow$	A point in $\mathbb{R}^2$	A point in $\mathbb{R}^2$	0
$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$	$\longrightarrow$	Whole of $\mathbb{R}^3$	Whole of $\mathbb{R}^3$	3 (full rank)
$\begin{pmatrix} 1 & 2 & 3 \\ 3 & 6 & 9 \end{pmatrix}$	$\longrightarrow$	A line in $\mathbb{R}^2$	A line in $\mathbb{R}^3$	1
$\begin{pmatrix} 1 & 2 & 5 & 1 \\ 3 & 4 & 6 & 7 \end{pmatrix}$	$\longrightarrow$	Whole of $\mathbb{R}^2$	A 2D-plane in $\mathbb{R}^4$	2 (Full row rank)

Matrices – Multiplying a Matrix with a Vector

- ▶ When we multiply a matrix with a vector, we get a transformed vector. It's convenient to think of transforming vector spaces instead of just a single vector. That way we can classify some very useful matrices based on their resulting transformation of vector spaces such as Rotation, Stretching, Shear, Projection etc....



$$\mathbf{A} = \begin{pmatrix} 5 & 2 \\ 1 & 3 \end{pmatrix} \quad \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 5 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$



Row Picture (dot product)

Column Picture (Column space)

(**b** must be in **A**'s column space)

$\mathbf{b} = \begin{pmatrix} 5x + 2y \\ x + 3y \end{pmatrix}$

 $\mathbf{b} = x \begin{pmatrix} 5 \\ 1 \end{pmatrix} + y \begin{pmatrix} 2 \\ 3 \end{pmatrix}$

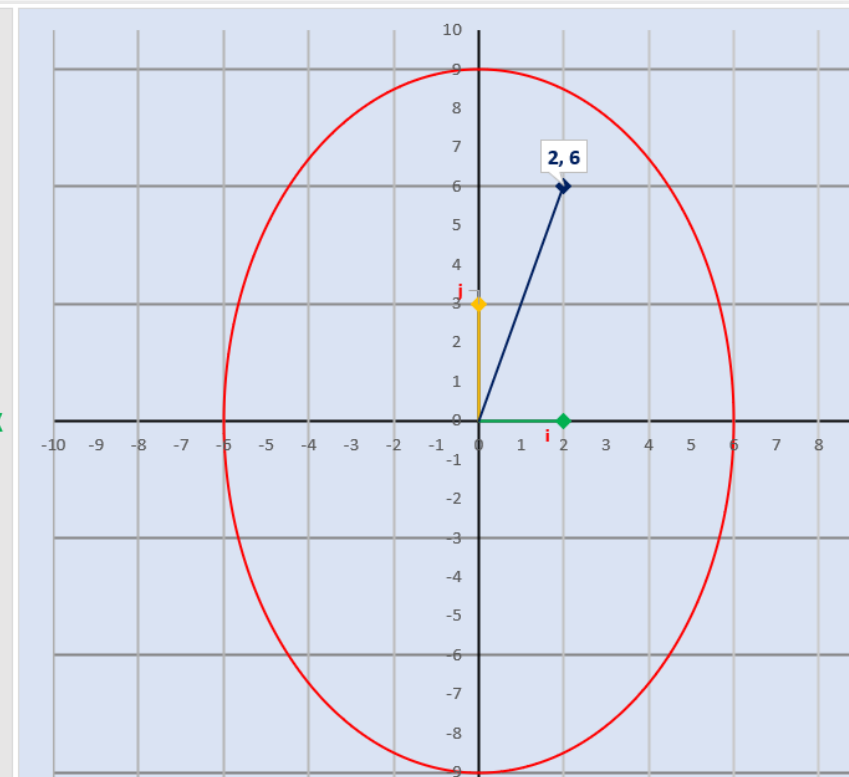
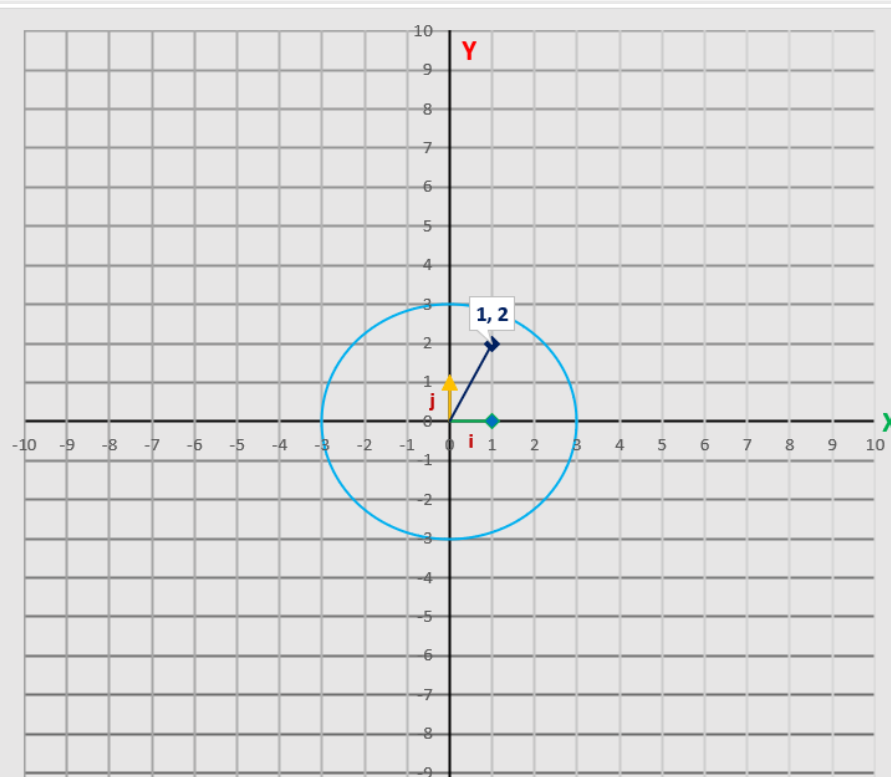
Vector Transformation - Stretching

► Here are some special transformations of $\mathbf{x}$ when we multiply $A\mathbf{x}$



$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$





Vector Transformation with Orthogonal Matrices

► **A** is an orthogonal matrix if $A^{-1} = A^T$ or $AA^T = I$

- I. Denoted as **Q**
- II. Columns and Rows are orthogonal
- III. Size of a vector remains the same with linear transformation
- IV. Angle between two vectors remain the same with linear transformation (combining III and IV, we can say dot product remains intact after transformation)
- V. Determinant is either +1 or -1

Identity matrix (no change to x) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Permutation matrix (swaps the components) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

Rotation matrix (rotates by a given angle) $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$

Reflection matrix (reflection along a vector) $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

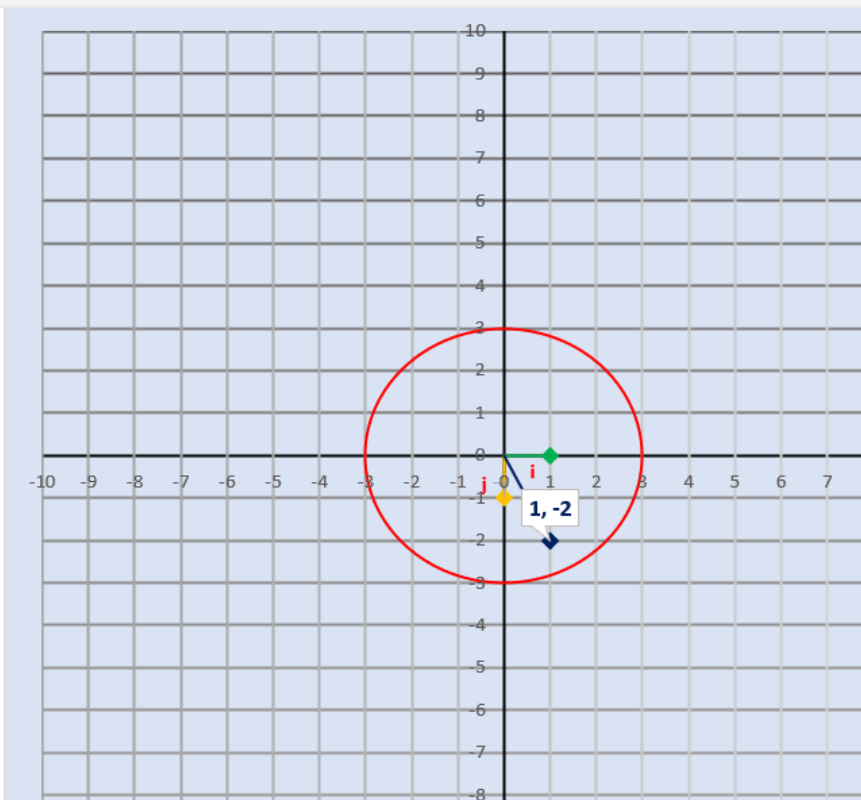
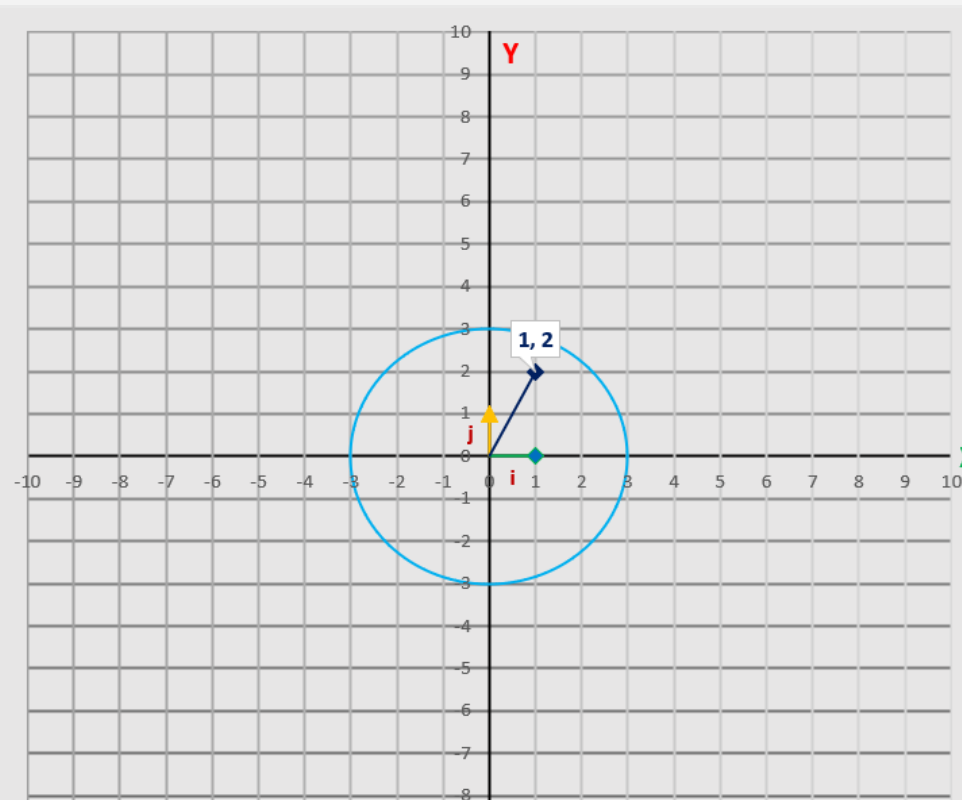
Vector Transformation - Reflection

► Here are some special transformations of $\mathbf{x}$ when we multiply $\mathbf{Ax}$



$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



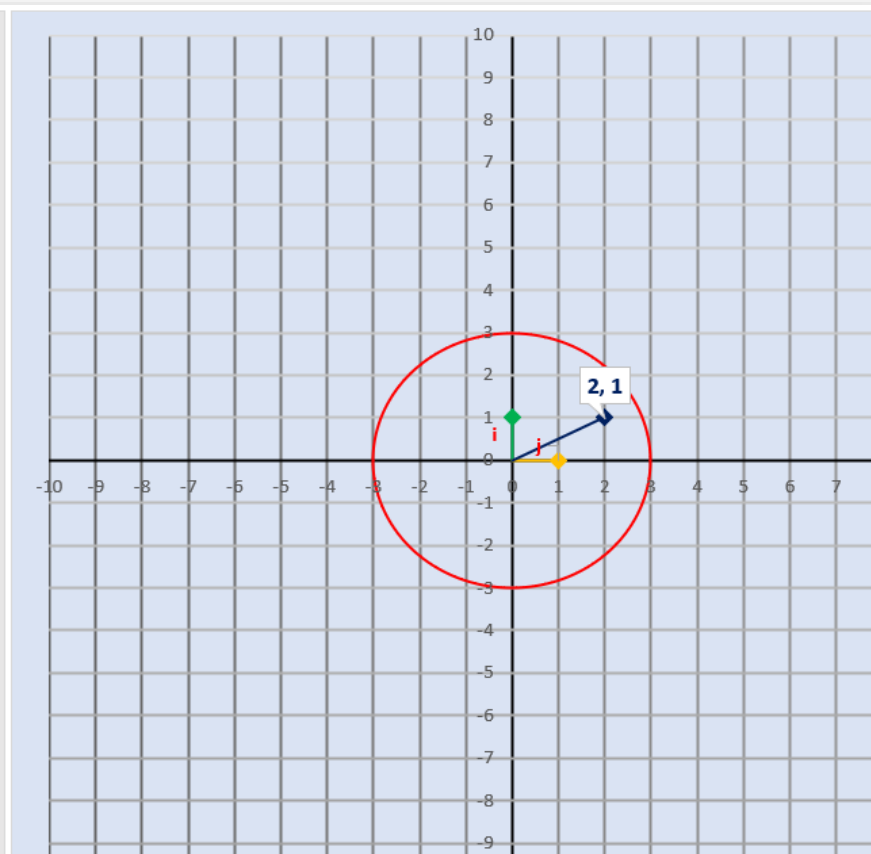
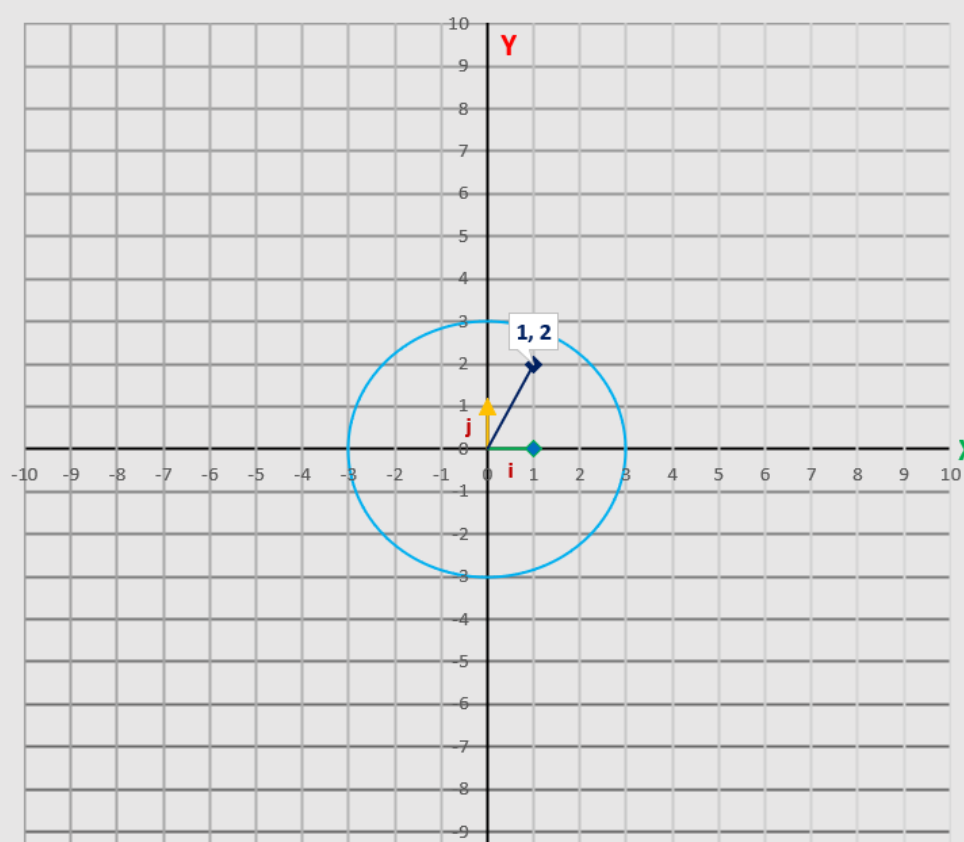
Vector Transformation - Permutation

► Here are some special transformations of $\mathbf{x}$ when we multiply $\mathbf{Ax}$



$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$



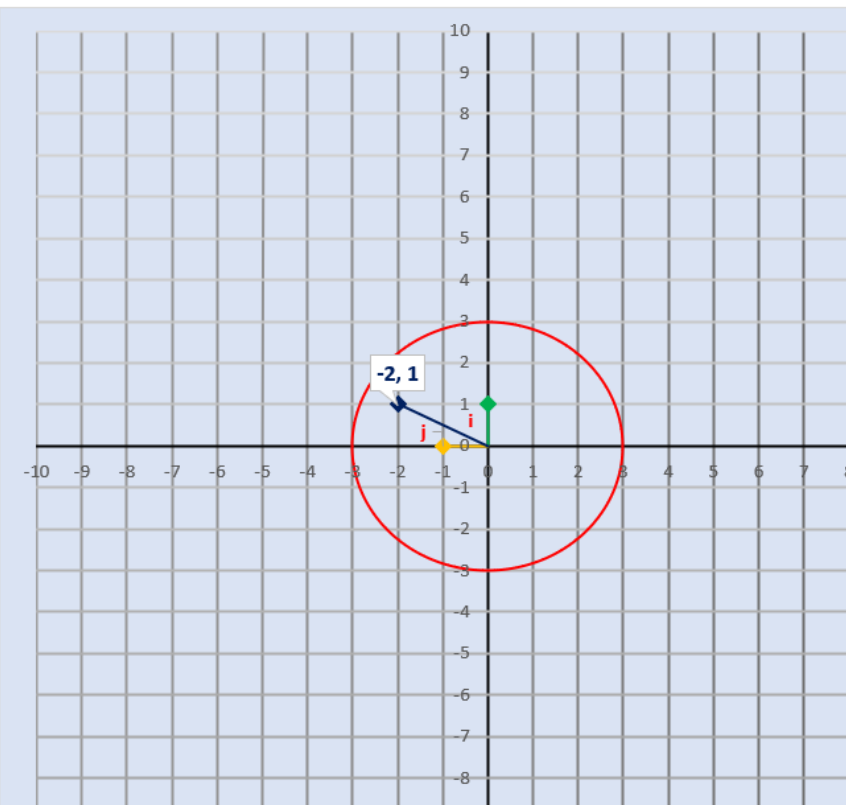
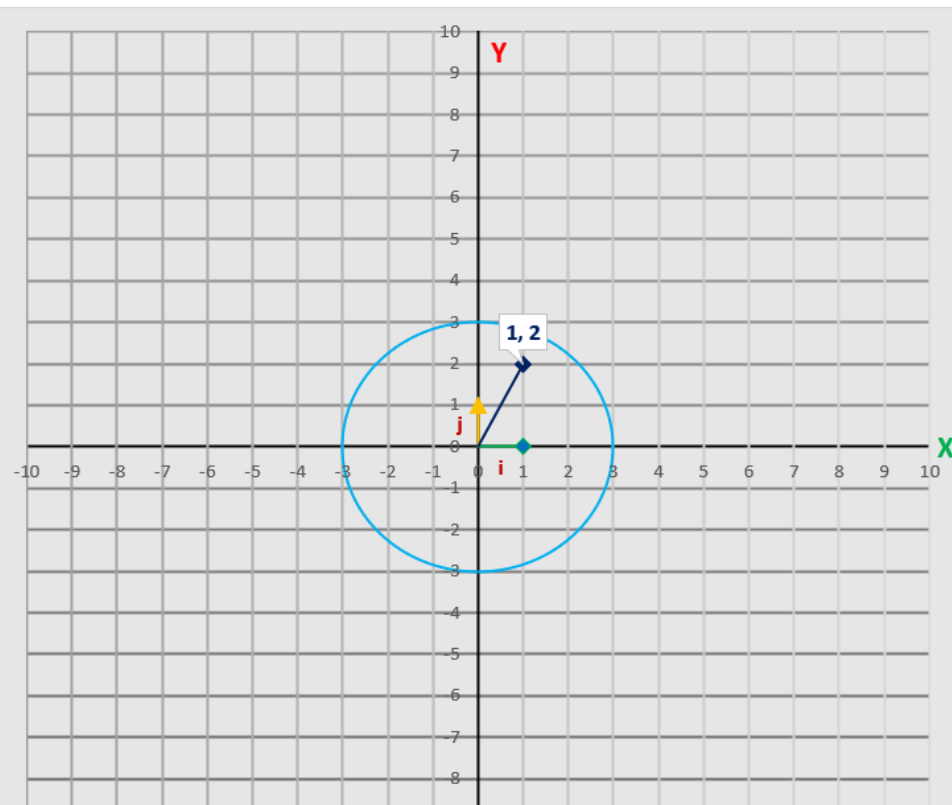
Vector Transformation - Rotation

► Here are some special transformations of $\mathbf{x}$ when we multiply $\mathbf{Ax}$



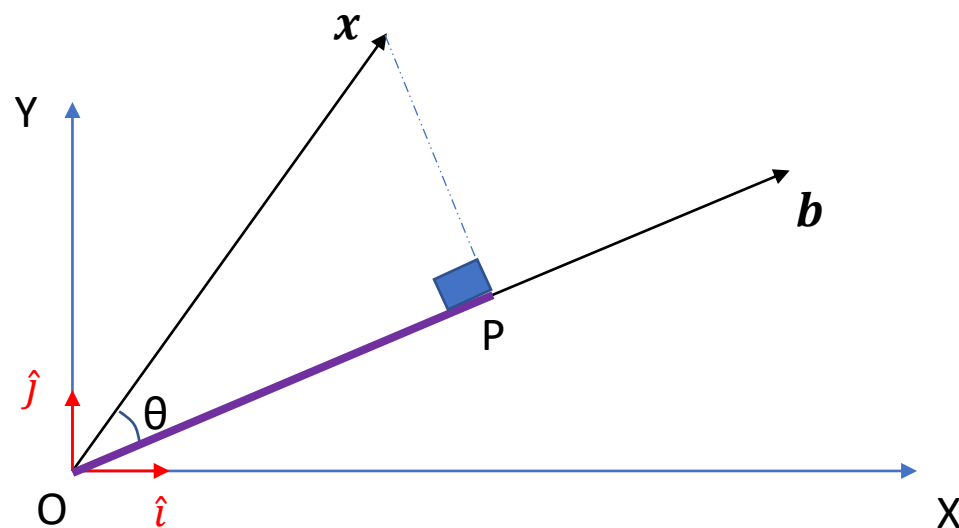
$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 0.00 & -1.00 \\ 1.00 & 0.00 \end{pmatrix}$$



Projection Matrix on a line sub-space

- $\mathbf{V}$ is the vector space and $\mathbf{U}$ is a subspace of V ($U \subseteq V$). A linear mapping $P: V \rightarrow U$ is called a projection if $P^2 = P$



$$\text{Projection Matrix : } P = \frac{bb^T}{|b|^2}$$

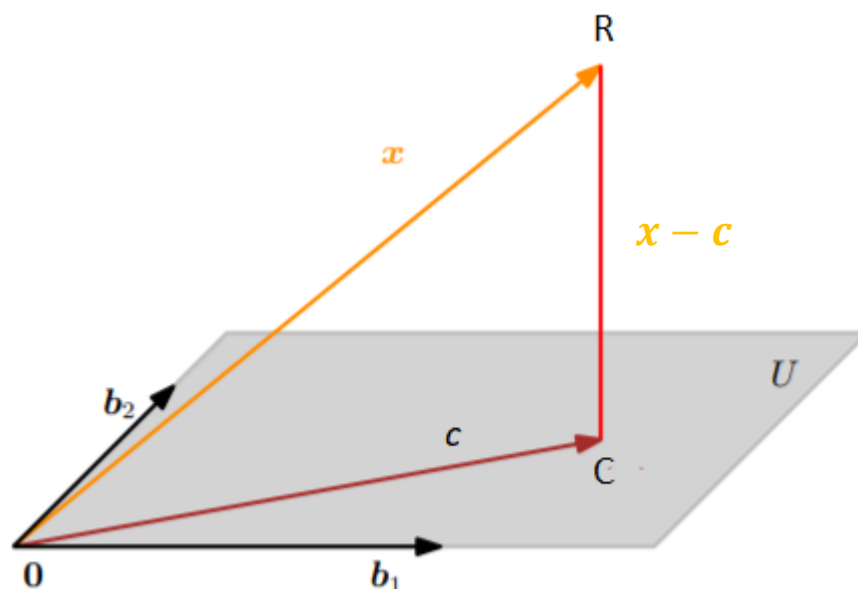
$$P\mathbf{x} = \overrightarrow{OP}$$

$\mathbf{V} = XY$ plane and $U = \mathbf{b}$ (a line) , Projection of any vector $\mathbf{x}$ in the entire XY plane on a given vector $\mathbf{b}$ falls on $\mathbf{b}$

This is a linear transformation and can be captured by the Projection matrix ($\mathbf{P}$)

What is column space and rank of $\mathbf{P}$?

Projection into general sub-spaces



- I. $\mathbf{c}$ is the projection of $\mathbf{x}$ on plane U
- II. $\mathbf{c}$ will be different for different $\mathbf{x}$ (**trick is to find $\mathbf{c}$**)
- III. $\mathbf{c}$ would be some linear combination of $\mathbf{b}_1$ and $\mathbf{b}_2$ which define the plane (say $\mathbf{c} = \mathbf{B}\boldsymbol{\lambda}$, where $\mathbf{B}$ is the matrix with $\mathbf{b}_1$ and $\mathbf{b}_2$ as columns and $\boldsymbol{\lambda}$ is the vector for the linear combination of $\mathbf{b}_1$ and $\mathbf{b}_2$)
- IV. But $(\mathbf{x} - \mathbf{c})$ will be perpendicular to both $\mathbf{b}_1$ and $\mathbf{b}_2$

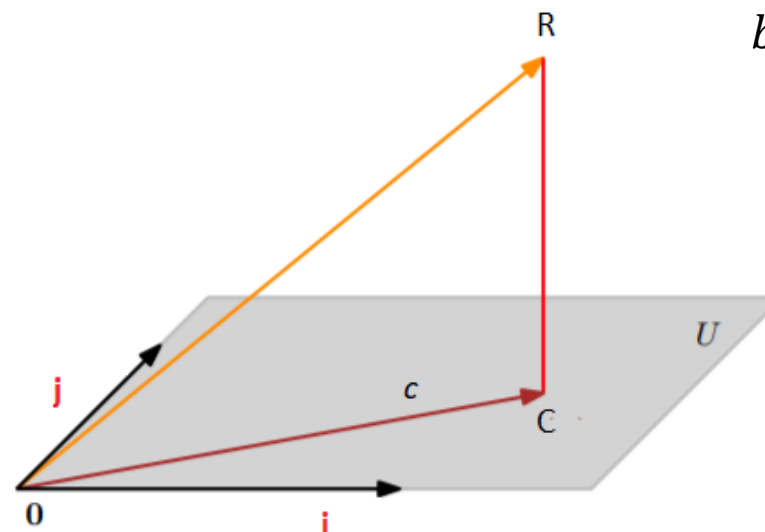
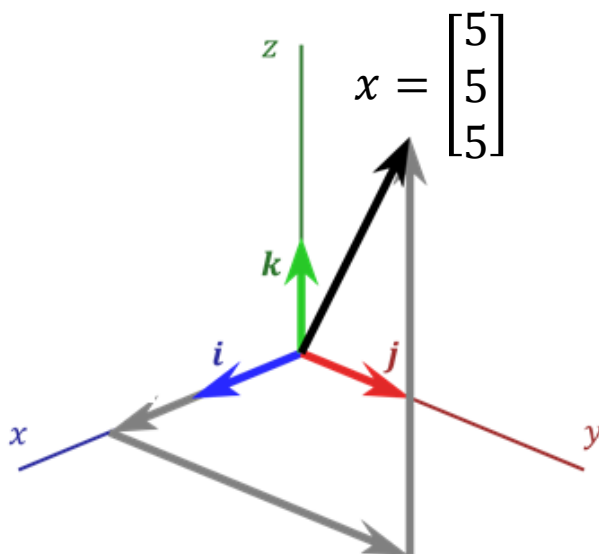
Results

$$\mathbf{B}^T \mathbf{B} \boldsymbol{\lambda} = \mathbf{B}^T \mathbf{x} \quad (\text{normal equation})$$

$$\boldsymbol{\lambda} = (\mathbf{B}^T \mathbf{B})^{-1} \mathbf{B}^T \mathbf{x} \quad (\text{coefficients})$$

$$(\mathbf{B}^T \mathbf{B})^{-1} \mathbf{B}^T \quad (\text{pseudo-inverse of } \mathbf{B})$$

Projection into general sub-spaces - Example



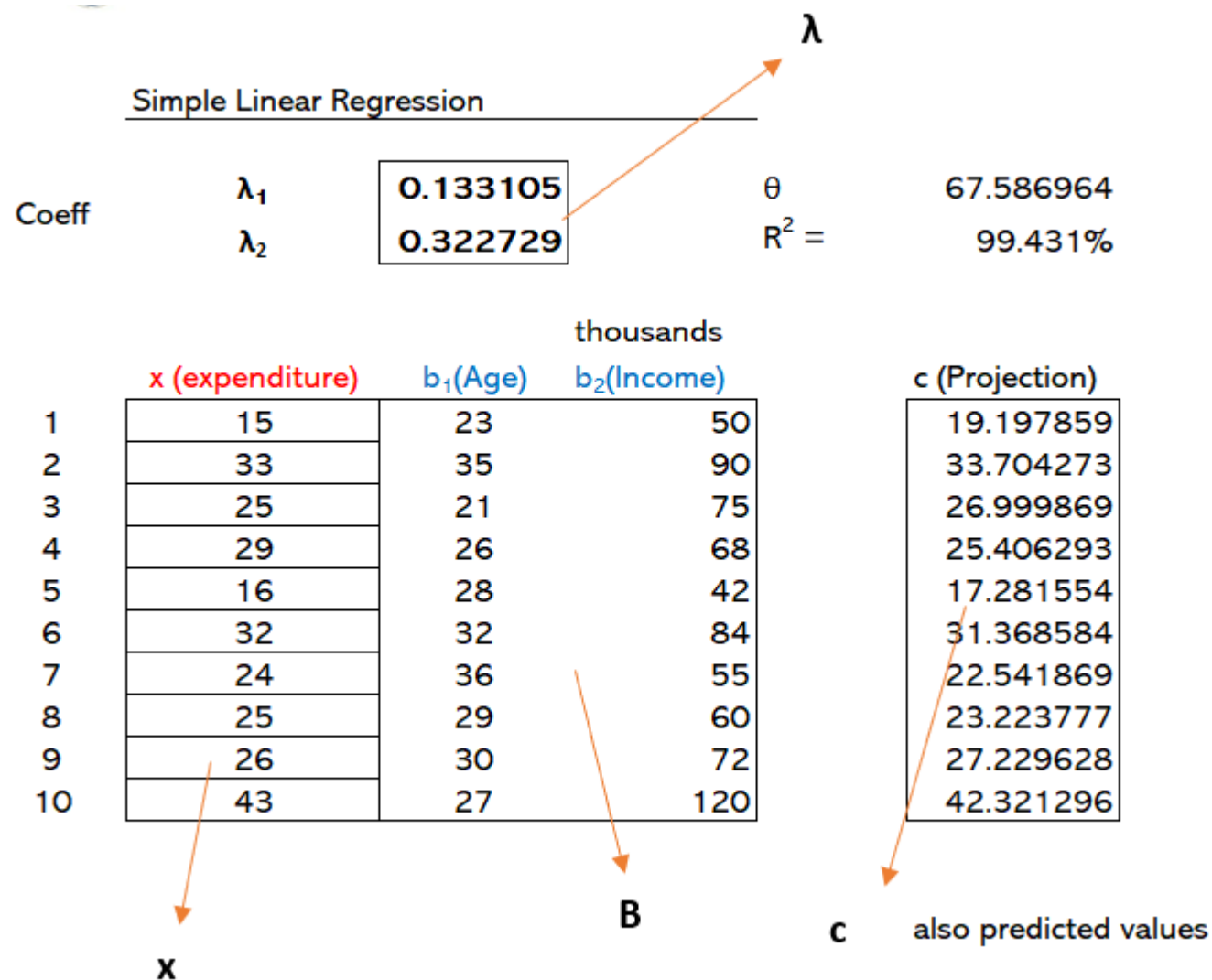
$$b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad b_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Find Projection matrix P

Find λ

Find c

Regression by Projection



Task

Predict expenditure using Age and Income as input

$Z = f(b_1, b_2)$... an exact linear formulation does not exist

Projection helps us to seek solutions which are not exact

Matrices – Solving a system of linear equations



- A system of linear equations can be solved using Matrix Algebra. Say we have 2 equations with 2 unknowns

$$5x + 2y = 9$$

$$\mathbf{Ax} = \mathbf{b}$$

$$x + 3y = 7$$

$$\mathbf{A} \quad \mathbf{x} \quad \mathbf{b}$$

$$\begin{pmatrix} 5 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 9 \\ 7 \end{pmatrix}$$

Multiply $\mathbf{A}^{-1}$ on both sides to get

$$\mathbf{A}^{-1}\mathbf{Ax} = \mathbf{A}^{-1}\mathbf{b}$$

$$\mathbf{Ix} = \mathbf{A}^{-1}\mathbf{b}$$

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

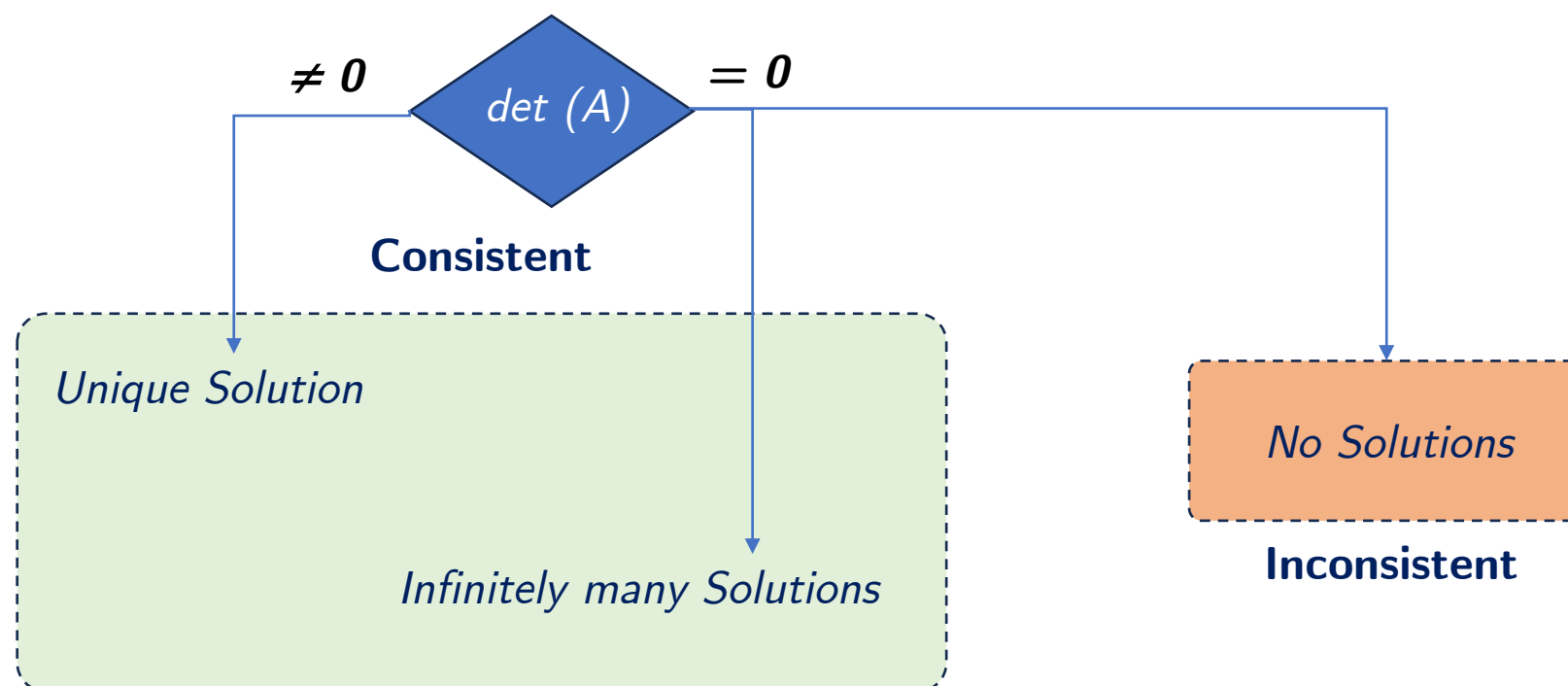
$$\mathbf{A}^{-1} = \begin{pmatrix} 0.2308 & -0.1538 \\ -0.0769 & 0.3846 \end{pmatrix} \quad |\mathbf{A}| = \mathbf{13} \neq \mathbf{0}$$

$$\text{Solution : } \mathbf{x} = \begin{pmatrix} 0.2308 & -0.1538 \\ -0.0769 & 0.3846 \end{pmatrix} \begin{pmatrix} 9 \\ 7 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

if $\mathbf{A}^{-1}$ exists ($\det(\mathbf{A}) \neq 0$), solution exists for all $\mathbf{b}$
 $\det(\mathbf{A}) \neq 0$ means matrix is *full rank*

Matrices – Solving a system of linear equations

- ▶ A system of linear equations is called consistent if there is at least one solution and inconsistent if there are no solutions
- ▶ If $\det(A) = 0$, then the system is either consistent with infinitely many solutions or inconsistent (no solutions)



Matrices – Solving a system of linear equations



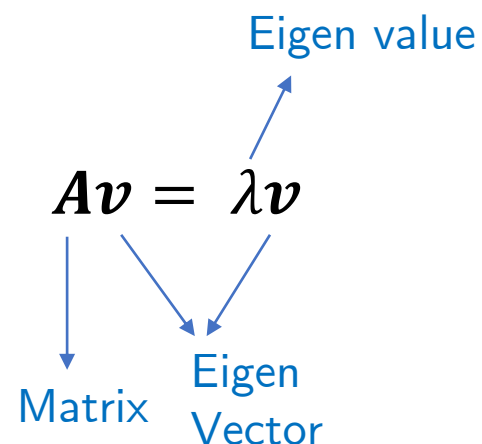
Exercise – Find the solution to the following system of equations.

a) $x + 2y = 8$
 $2x + 4y = 16$

b) $x + 2y = 8$
 $2x + 4y = 15$

Matrices – Eigen Value and Eigen Vector

- ▶ We saw that matrix multiplied with a vector performs a linear transformation within the column space of the matrix.
- ▶ The vector(s) which only get stretched (scaled) and do not change direction is(are) called 'eigen vector(s)' of the matrix.
- ▶ The scalar by which each eigen vector gets stretched is the corresponding eigen value.
- ▶ Mathematically,

$$A\mathbf{v} = \lambda\mathbf{v}$$


The diagram illustrates the equation $A\mathbf{v} = \lambda\mathbf{v}$. Arrows point from the terms to their definitions: A points to "Matrix", $\mathbf{v}$ points to "Eigen Vector", and λ points to "Eigen value".



Characteristic Polynomial

$$A\mathbf{v} = \lambda\mathbf{v}$$

$$A\mathbf{v} = \lambda I\mathbf{v}$$

$$(A - \lambda I)\mathbf{v} = \mathbf{0} \quad (\text{Homogeneous equation})$$

- ▶ This system always has a solution $\mathbf{v} = \mathbf{0}$ (trivial solution)
- ▶ If the above solution is not unique, there will be infinitely many solutions
- ▶ We are interested to see the structure of these infinite solutions

But for the system to have infinitely many solutions we need $|(A - \lambda I)| = 0$

$$|(A - \lambda I)| = 0$$



Characteristic equation of A

Eigen Value and Eigen Vector - Example

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad A\mathbf{v} = \lambda\mathbf{v} \quad (A - \lambda I)\mathbf{v} = \mathbf{0}$$

$$|(A - \lambda I)| = 0 \quad (\text{Characteristic equation of } A)$$

$$(A - \lambda I) = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{pmatrix}$$

$$|(A - \lambda I)| = (2 - \lambda)^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0 \quad \lambda = 1, 3 \quad (\text{Eigen values})$$

$$\begin{array}{cc} \lambda = 1 & \lambda = 3 \\ \swarrow & \searrow \\ \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} & \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \\ \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} & \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{array}$$

Properties of Eigen Vectors and Eigen Values

- ▶ Trace of A (sum of diagonal entries) = sum of eigen values $\text{tr}(A) = \sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i = \lambda_1 + \lambda_2 + \cdots + \lambda_n$
- ▶ Determinant of A = product of eigen values $\det(A) = \prod_{i=1}^n \lambda_i = \lambda_1 \lambda_2 \cdots \lambda_n$
- ▶ Eigen values of A^k = k^{th} power of eigen values $\lambda_1^k, \dots, \lambda_n^k$
- ▶ A is invertible only when all eigen values are non-zero
- ▶ Eigen values of A^{-1} are reciprocal of eigen values of A $\frac{1}{\lambda_1}, \dots, \frac{1}{\lambda_n}$
- ▶ Eigen vectors of a symmetric matrix are orthogonal
- ▶ Eigen values of a positive definite matrix are all >0
- ▶ Eigen values of a positive semi-definite matrix are ≥ 0

Matrix Decomposition - LU

Factors a matrix into a **lower triangular** matrix and an **upper triangular** matrix.

$$A = LU \quad \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} \ell_{11} & 0 & 0 \\ \ell_{21} & \ell_{22} & 0 \\ \ell_{31} & \ell_{32} & \ell_{33} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

Computers use LU decomposition to solve system of linear equations as Inverse is easier.

$$A\mathbf{x} = \mathbf{b} \quad LU\mathbf{x} = \mathbf{b} \quad U\mathbf{x} = L^{-1}\mathbf{b}$$

Matrix Decomposition – LU (2x2 example)

2 x 2 case

$$\overset{A}{\begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix}} \xrightarrow{-4R_1+R_2} \overset{U}{\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}}$$

$$\overset{E}{\begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}} \overset{A}{\begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix}} = \overset{U}{\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}}$$

$$\overset{A}{\begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix}} = \overset{E^{-1}}{\begin{pmatrix} 1 & 0 \\ -4 & 1 \end{pmatrix}^{-1}} \overset{U}{\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}}$$

$$\overset{A}{\begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix}} = \overset{L}{\begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}} \overset{U}{\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}}$$

LU Decomposition

$$\overset{A}{\begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix}} = \overset{L}{\begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}} \overset{D}{\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}} \overset{U}{\begin{pmatrix} 1 & 1/2 \\ 0 & 1 \end{pmatrix}}$$

LDU Decomposition

D when pre multiplied, scales the rows
D when post multiplied scales the columns

Matrix Decomposition – LU (3x3 example)

3 x 3 case

$$\begin{matrix} \text{A} \\ \left[\begin{array}{ccc} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{array} \right] \end{matrix} \xrightarrow[\substack{R_2 \\ -R_1 + R_2}]{\substack{R_2 \\ -R_1 + R_2}} \begin{matrix} \\ \left[\begin{array}{ccc} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 2 & 3 & 1 \end{array} \right] \end{matrix}$$

$$\left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \quad E_{21}$$

$$\text{A} = \text{L} \text{U} \quad \left[\begin{array}{ccc} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{array} \right] = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/3 & 1 \end{array} \right] \left[\begin{array}{ccc} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 13/3 \end{array} \right]$$

$$\begin{matrix} \\ \left[\begin{array}{ccc} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 2 & 3 & 1 \end{array} \right] \end{matrix} \xrightarrow[\substack{R_3 \\ -2R_1 + R_3}]{\substack{R_3 \\ -2R_1 + R_3}} \begin{matrix} \\ \left[\begin{array}{ccc} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 1 & 3 \end{array} \right] \end{matrix}$$

$$\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{array} \right] \quad E_{31}$$

$$\text{A} = \text{L} \text{D} \text{U} \quad \left[\begin{array}{ccc} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{array} \right] = \left[\begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/3 & 1 \end{array} \right] \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 13/3 \end{array} \right] \left[\begin{array}{ccc} 1 & 1 & -1 \\ 0 & 1 & -4/3 \\ 0 & 0 & 1 \end{array} \right]$$

$$\begin{matrix} \\ \left[\begin{array}{ccc} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 1 & 3 \end{array} \right] \end{matrix} \xrightarrow[\substack{R_3 \\ \frac{1}{3}R_2 + R_3}]{\substack{R_3 \\ \frac{1}{3}R_2 + R_3}} \begin{matrix} \\ \left[\begin{array}{ccc} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 13/3 \end{array} \right] \end{matrix}$$

$$\left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1/3 & 1 \end{array} \right] \quad E_{32}$$

$$\text{A} = \text{LU}$$

$$\text{L} = E_{21}^{-1} E_{31}^{-1} E_{32}^{-1}$$

$$\left[\begin{array}{ccc} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -1/3 & 1 \end{array} \right]$$

$$\begin{matrix} E_{32} & E_{31} & E_{21} & \text{A} & \text{U} \\ \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1/3 & 1 \end{array} \right] \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{array} \right] \left[\begin{array}{ccc} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right] \left[\begin{array}{ccc} 1 & 1 & -1 \\ 1 & -2 & 3 \\ 2 & 3 & 1 \end{array} \right] = \left[\begin{array}{ccc} 1 & 1 & -1 \\ 0 & -3 & 4 \\ 0 & 0 & 13/3 \end{array} \right] \end{matrix}$$

Matrix Decomposition – Cholesky Decomposition

A is a positive definite matrix if $\mathbf{w}^T \mathbf{A} \mathbf{w} > 0$ $\mathbf{w}$ is a non-zero vector

A is a positive semi definite matrix if $\mathbf{w}^T \mathbf{A} \mathbf{w} \geq 0$

A is a symmetric positive definite matrix, then $A = LL^T$

$$\mathbf{A} = \mathbf{L} \mathbf{D} \mathbf{U}$$

$$\begin{bmatrix} 1 & 0.5 & 0.2 \\ 0.5 & 1 & -0.3 \\ 0.2 & -0.3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 1 & 0 \\ 0.2 & -0.53 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.75 & 0 \\ 0 & 0 & 0.746 \end{bmatrix} \begin{bmatrix} 1 & 0.5 & 0.2 \\ 0 & 1 & -0.53 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0.5 & 0.2 \\ 0.5 & 1 & -0.3 \\ 0.2 & -0.3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 1 & 0 \\ 0.2 & -0.533 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & 0 \\ 0 & 0 & 0.8641 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & 0 \\ 0 & 0 & 0.8641 \end{bmatrix} \begin{bmatrix} 1 & 0.5 & 0.2 \\ 0 & 1 & -0.53 \\ 0 & 0 & 1 \end{bmatrix} \quad A = LDU = (LD^{1/2})(D^{1/2}U)$$

$$\begin{bmatrix} 1 & 0.5 & 0.2 \\ 0.5 & 1 & -0.3 \\ 0.2 & -0.3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 0.866025 & 0 \\ 0.2 & -0.46188 & 0.8641 \end{bmatrix} \begin{bmatrix} 1 & 0.5 & 0.2 \\ 0 & 0.866025 & -0.46188 \\ 0 & 0 & 0.8641 \end{bmatrix} \quad A = LL^T$$

Application : Drawing normal randoms from joint normal distribution

Matrix Decomposition – Eigen Value Decomposition

$$A = PDP^{-1}$$

Where,

P contains eigen vectors of A as columns

D is the diagonal matrix with eigen values of A on the diagonal

This decomposition (diagonalization) is possible if **P** is full rank
If not, then **A** is called defective (does not have a full basis of EV)

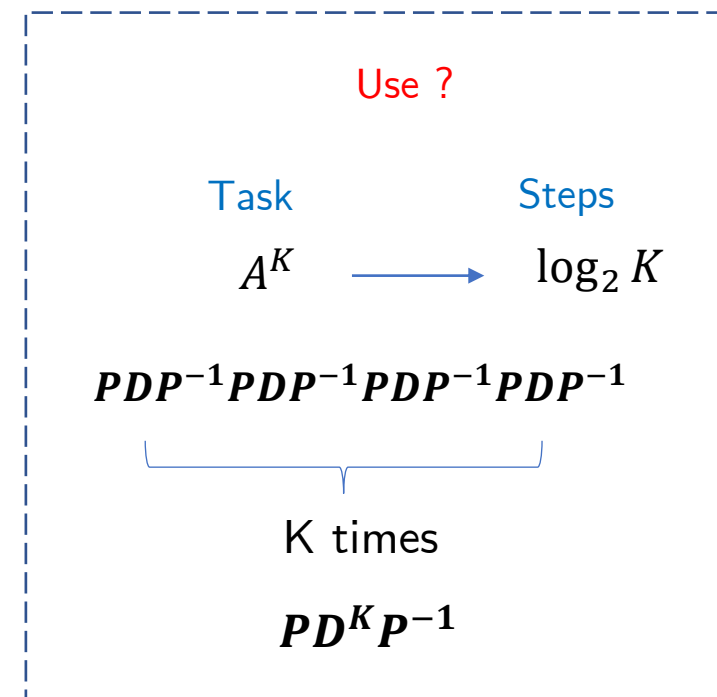
If A is symmetric, we can get $A = QDQ^T$, where Q is orthogonal

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \quad \lambda = 1, 3$$

$$A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{aligned} & \begin{matrix} \nearrow \\ \nearrow \end{matrix} A \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \longrightarrow A = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}^{-1} \end{aligned}$$



Matrix Decomposition – Singular Value Decomposition

$$A = U\Sigma V^T$$

$$V = \text{eigenvectors of } (A^T A) \quad \dots \text{normalized}$$

$$Av_i = \sigma_i u_i$$

$$U = \text{eigenvectors of } (AA^T) \quad \dots \text{normalized}$$

$$\Sigma = \text{diag}(\sigma_{ii}^2) \text{ of } AA^T \text{ arranged in descending order}$$

$$A = \begin{pmatrix} 2 & 2 \\ -1 & 1 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} 5 & 3 \\ 3 & 5 \end{pmatrix}$$

Eigen
value

8

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \xrightarrow{\text{normalize}} v_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$Av_1 = 2\sqrt{2} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Eigen
value

2

$$v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \xrightarrow{\text{normalize}} v_2 = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

$$Av_2 = \sqrt{2} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

U

Σ

V^T

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}$$

Matrix Decomposition – SVD (rank deficient)

$$\begin{aligned}
 A &= \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \\
 A^T A &= \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}
 \end{aligned}$$

$\swarrow$ Eigen value **10**
 $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \xrightarrow{\text{normalize}} v_1 = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$

$\searrow$ Eigen value **0**
 $v_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \xrightarrow{\text{normalize}} v_2 = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$

$$\begin{aligned}
 Av_1 &= \sqrt{10} \begin{pmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{pmatrix} \\
 Av_2 &= \sqrt{0} \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}
 \end{aligned}$$

$$A = \overset{\text{U}}{\begin{pmatrix} 2/\sqrt{5} & -1/\sqrt{2} \\ 1/\sqrt{5} & 1/\sqrt{2} \end{pmatrix}} \overset{\Sigma}{\begin{pmatrix} \sqrt{10} & 0 \\ 0 & 0 \end{pmatrix}} \overset{V^T}{\begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix}}$$

Matrix Decomposition – SVD (rectangular matrix)

$$A_{m \times n} = U_{m \times m} \Sigma_{n \times n} V^T_{n \times n}$$

$$(A^T A)_{n \times n} = V_{n \times n} (\Sigma^T \Sigma)_{n \times n} V^T_{n \times n}$$

$$(A A^T)_{m \times m} = U_{m \times m} (\Sigma \Sigma^T)_{m \times m} U^T_{m \times m}$$

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}_{2 \times 3}$$

$$A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

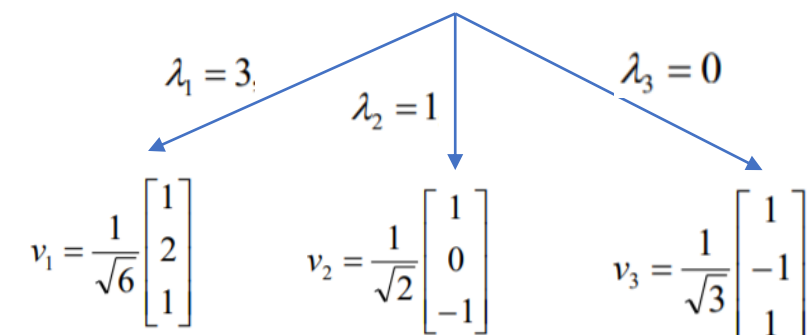


Diagram showing the eigenvalues of $A^T A$ and their corresponding eigenvectors v_1, v_2, v_3 .

- $\lambda_1 = 3$ with eigenvector $v_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$
- $\lambda_2 = 1$ with eigenvector $v_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$
- $\lambda_3 = 0$ with eigenvector $v_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$



Matrix Decomposition – SVD (Application)

$$A = U\Sigma V^T$$

$$A = \begin{bmatrix} u_1 & u_2 & u_3 & \dots & u_n \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_n & \\ & & & \ddots \end{bmatrix} \begin{bmatrix} v_1^T \\ \vdots \\ v_n^T \end{bmatrix}$$

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \dots \quad \text{With increase in terms accuracy will increase}$$

This gives a low rank approximation of A